

第十三章 取代酸 和 β -二羰基化合物

Substituted Carboxylic Acids
and β -Dicarbonyl Compounds

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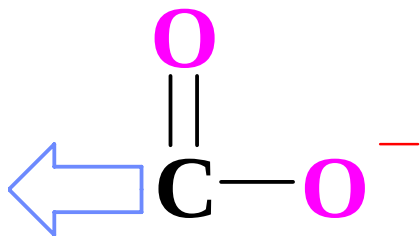
CONTENT

- 1 取代酸的酸性（自学）
- 2 羟基酸
- 3 羧基酸（氧代酸）
- 4 β -酮酸酯
- 5 β -二羰基化合物的反应

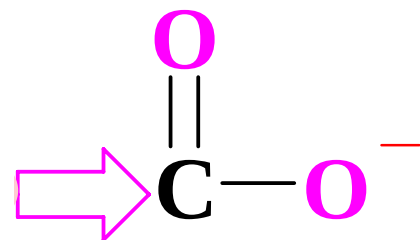
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13.1 取代酸的酸性（自学）

- 有利离解出质子的因素使酸性增强
- 不利离解出质子的因素使酸性减弱



提高羧酸根阴离子
稳定性的因素

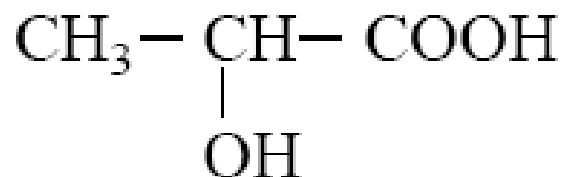


降低羧酸根阴离子
稳定性的因素

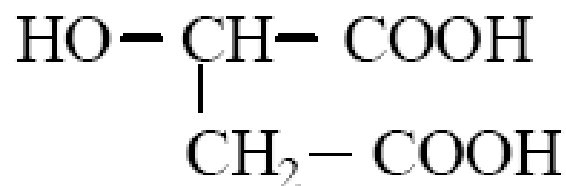
13.2 羟基酸

1. 羟基酸的命名和物理性质（自学）

许多羟基酸来自于动植物，常用俗名来表示



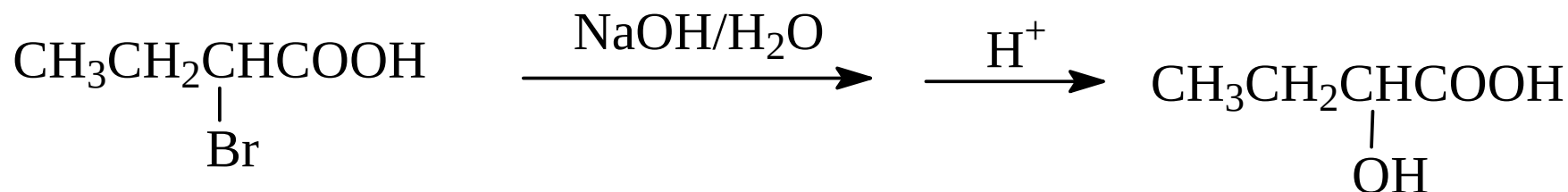
2-羟基丙酸（乳酸）



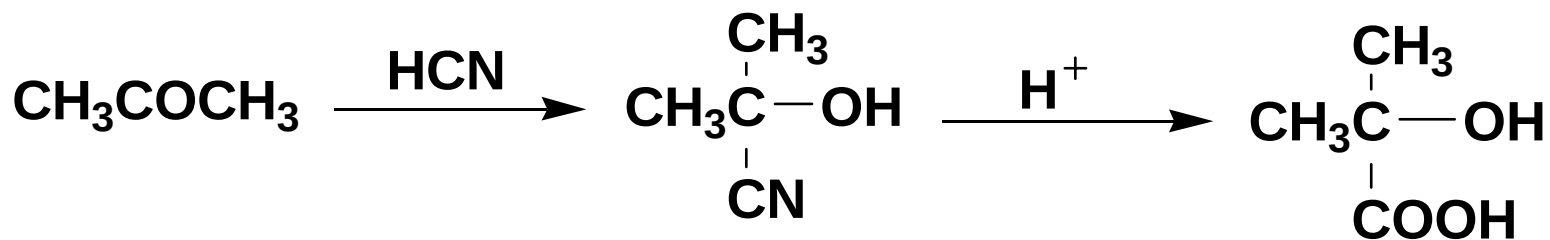
2-羟基丁二酸（苹果酸）

2. 羧酸的制备

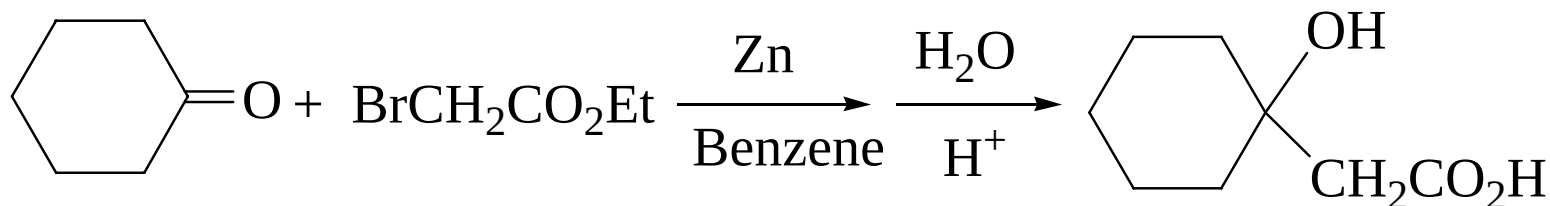
- 卤代酸水解



- 羟基腈水解

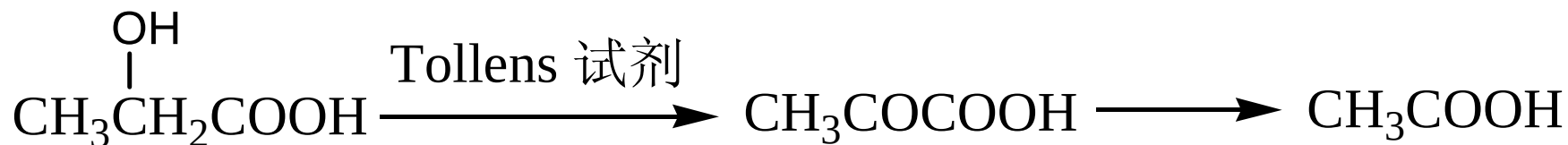


- Reformatsky反应



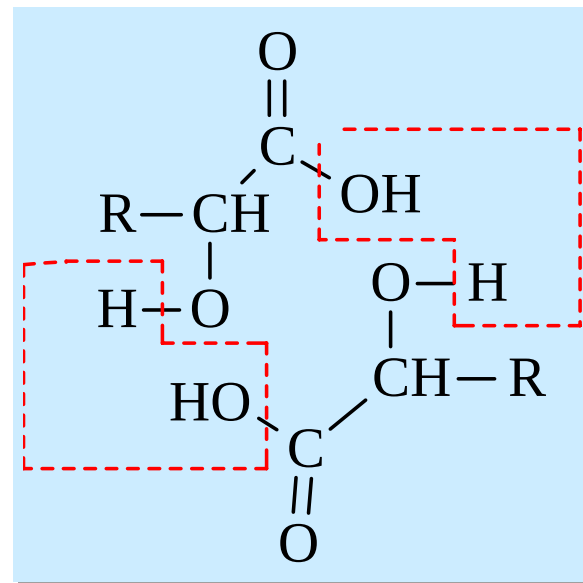
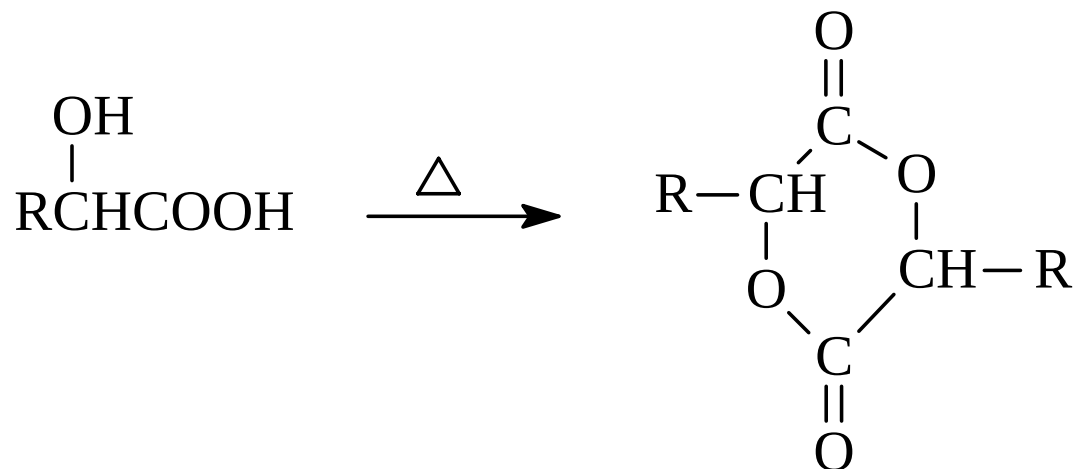
3. 羧基酸的反应

- 氧化反应

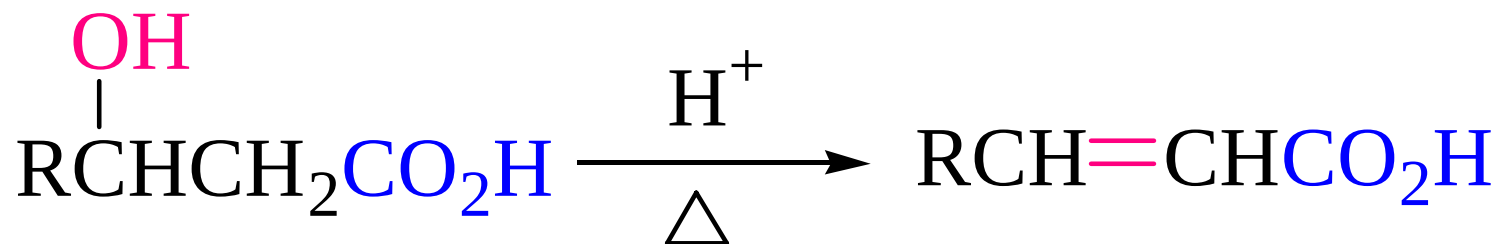


- 脱水反应

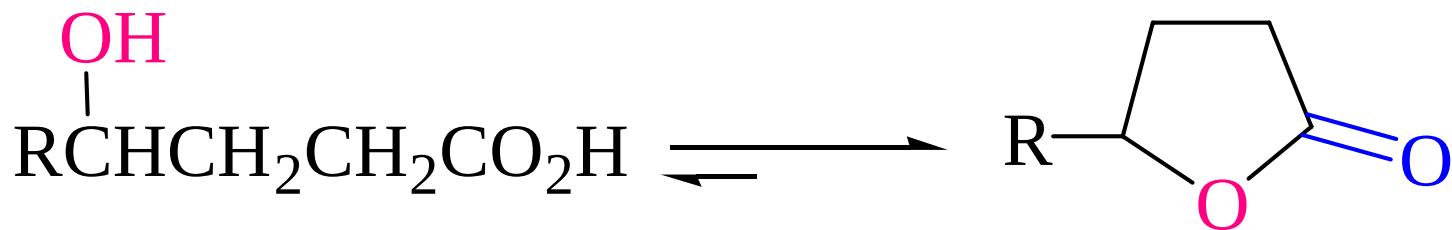
(1) α -羟基酸



(2) β -羟基酸



(3) γ -羟基酸

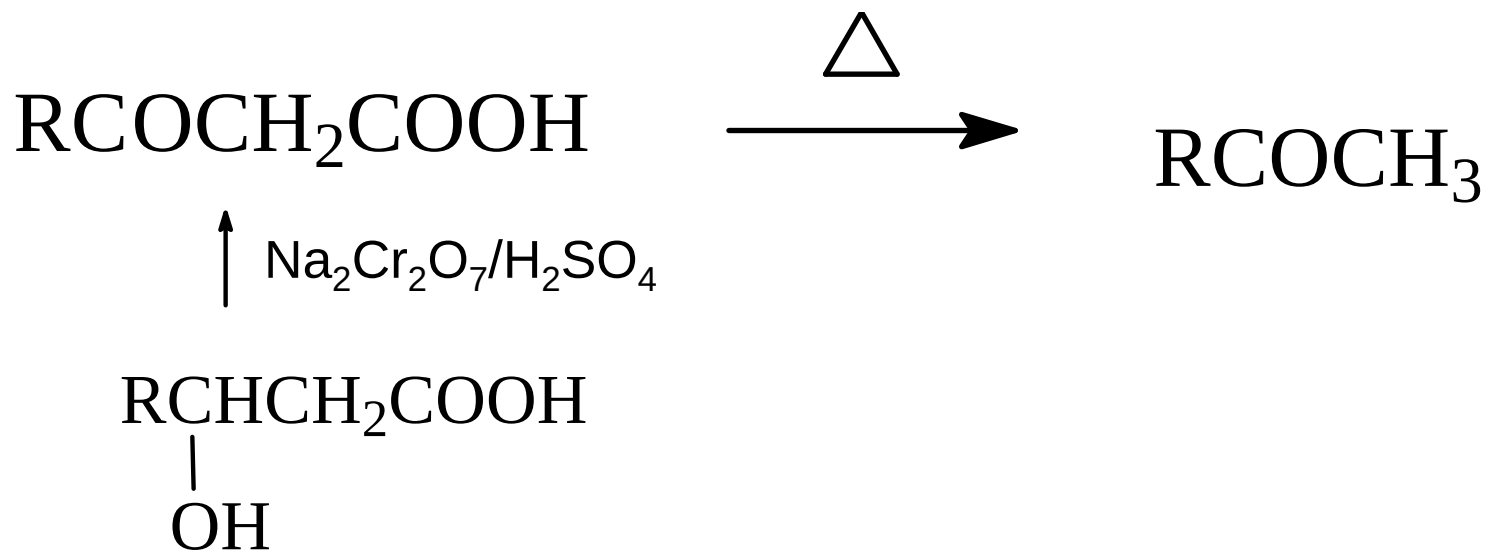


13.3 羧基酸（氧代酸）



3-氧代丁酸

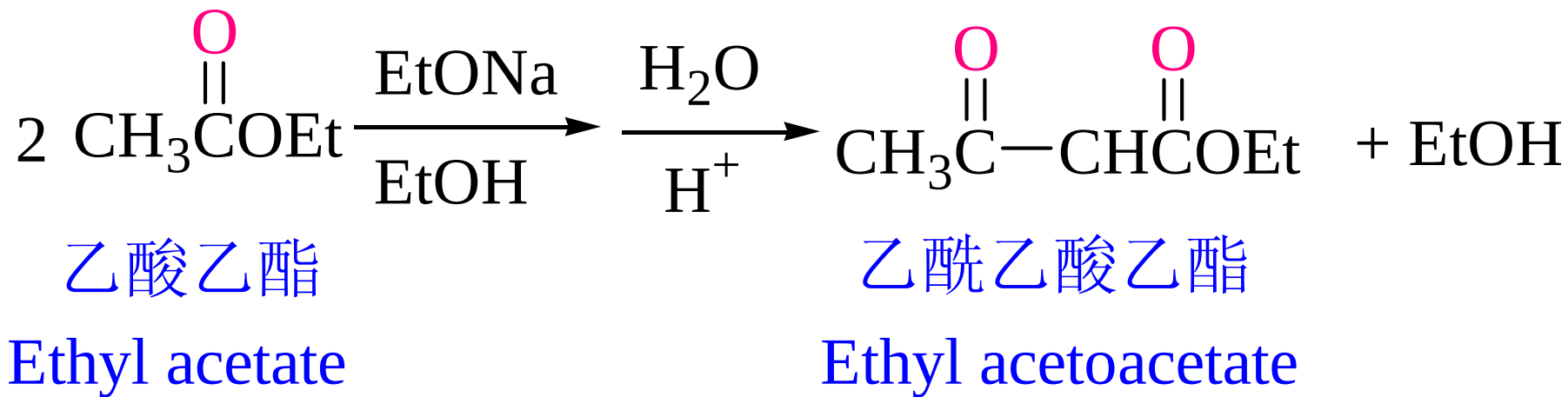
- 酸性大于羧酸
- α 或 β -酮酸受热易脱羧



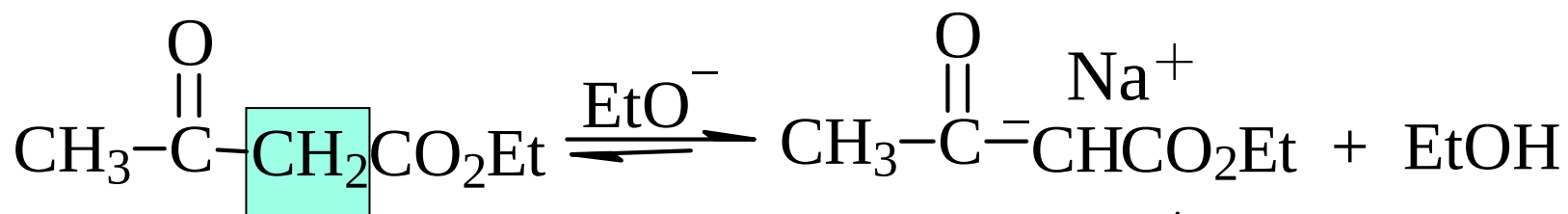
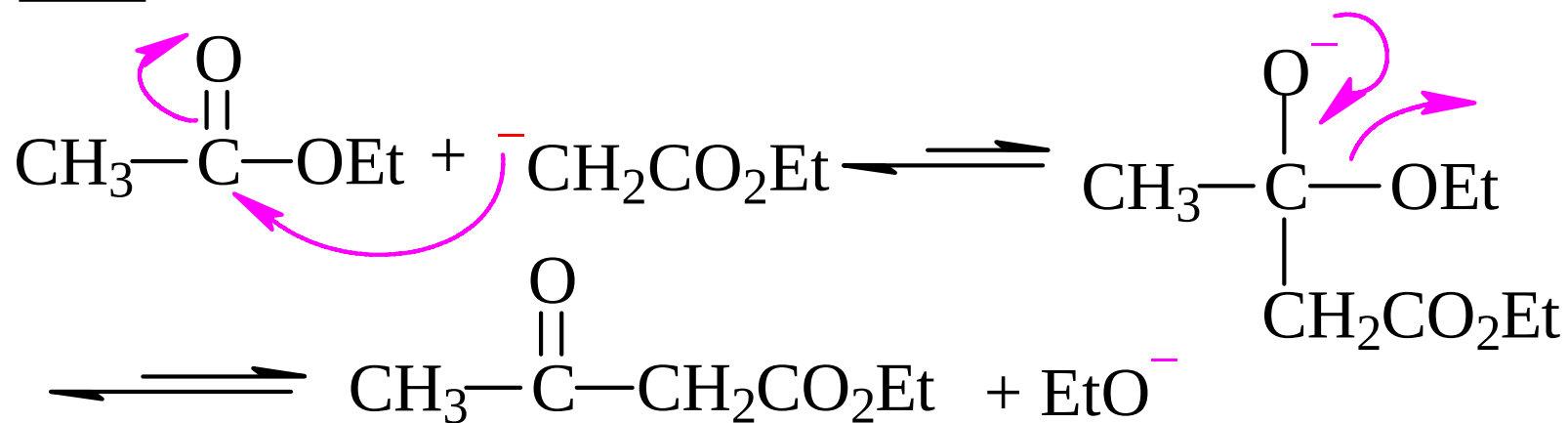
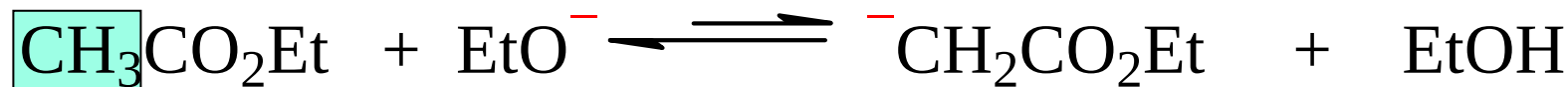
13.4 β -酮酸酯

1. β -酮酸酯的形成

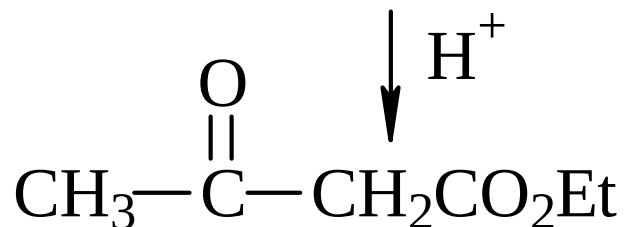
Claisen 酯缩合: 含 α -氢的酯在碱（醇钠）的作用下，两分子酯发生缩合反应，生成 β -羰基酸酯同时消去一分子醇



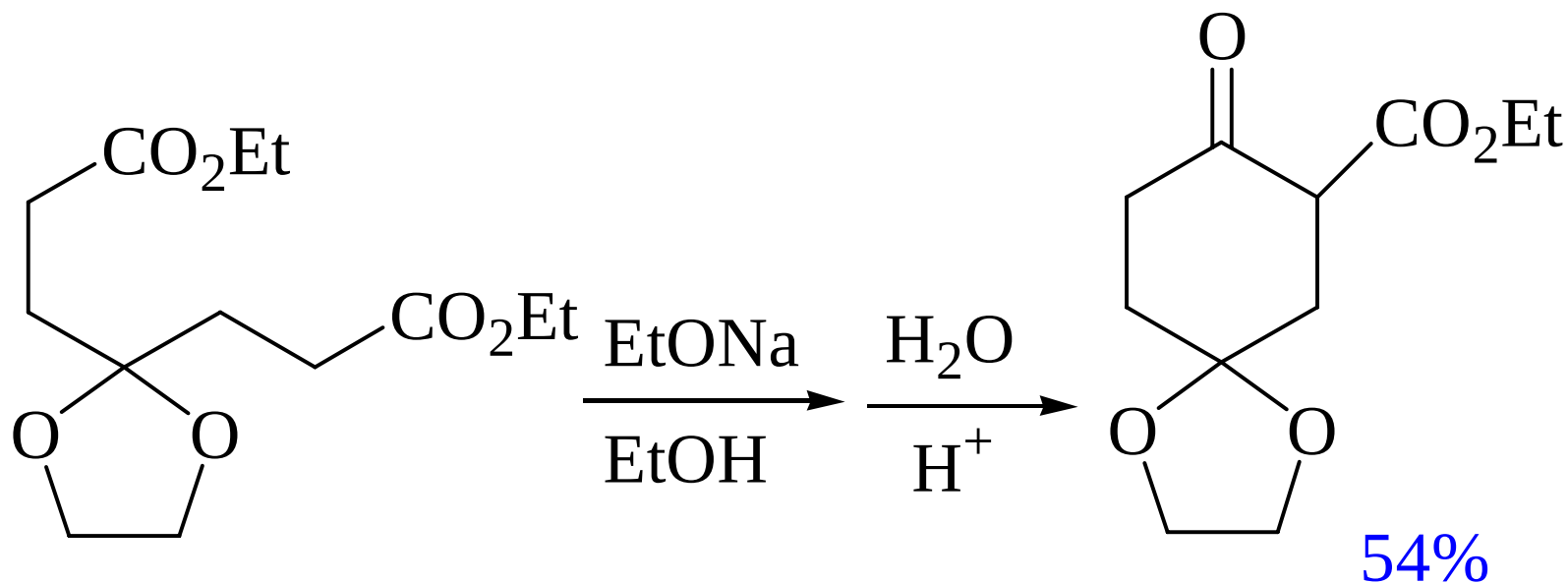
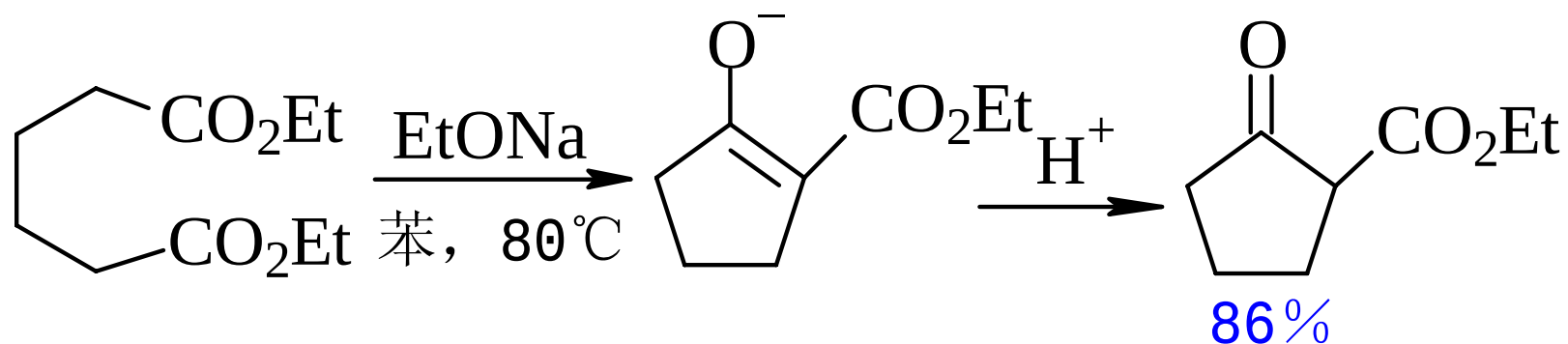
pKa 24.5



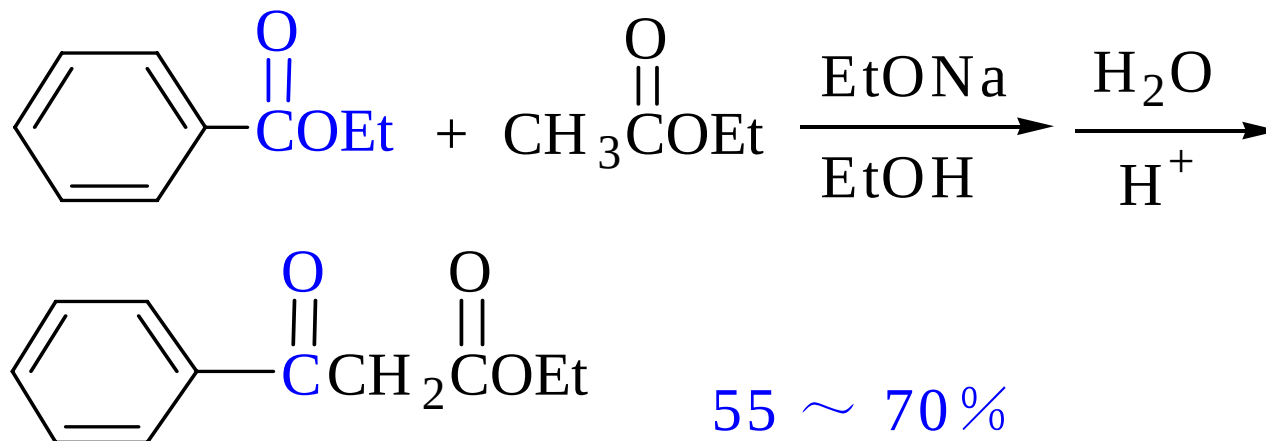
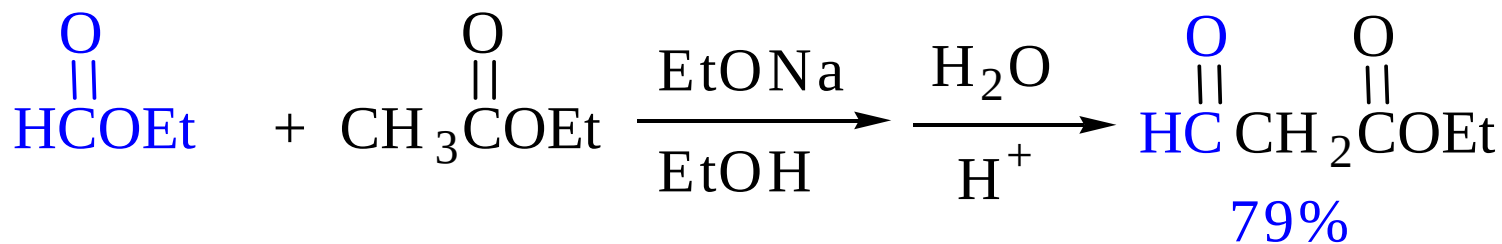
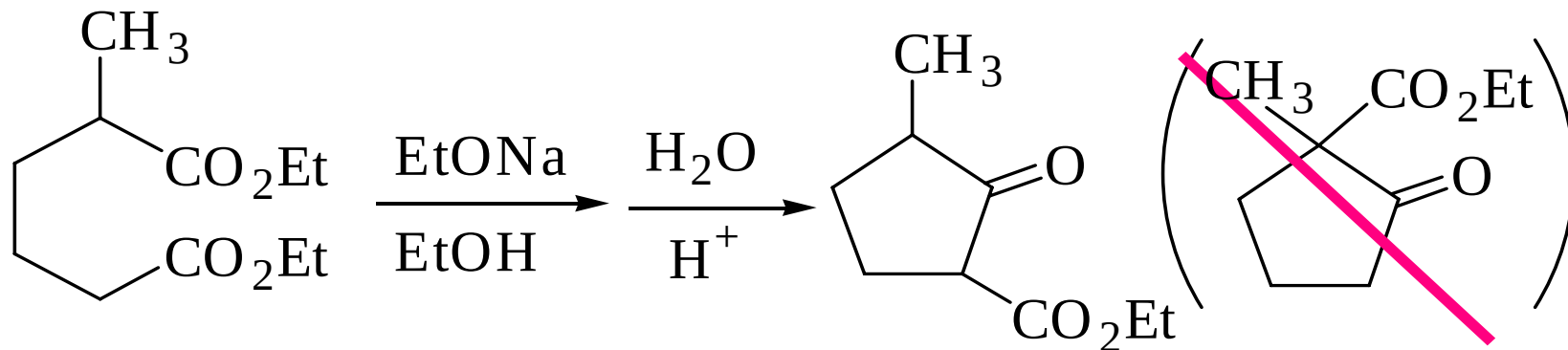
pKa ~11



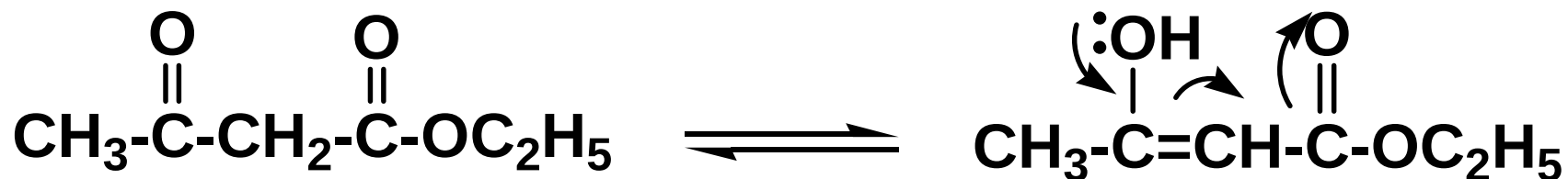
Dieckmann反应



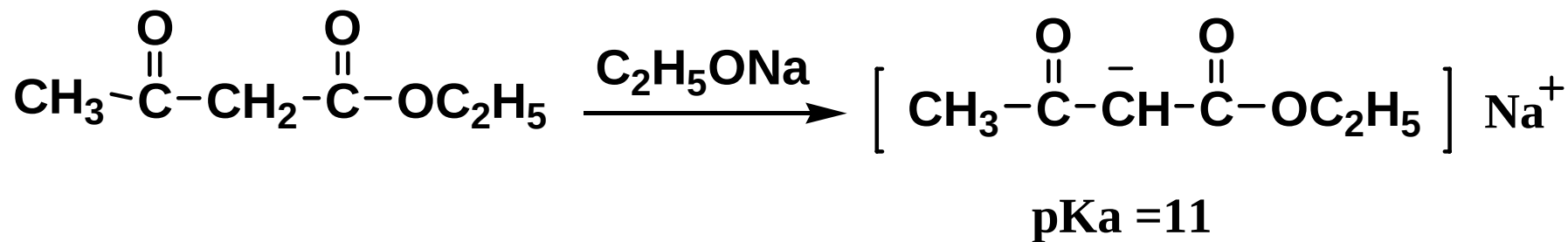
不对称二酯和混合酯的缩合



2. β -酮酸酯的互变异构现象



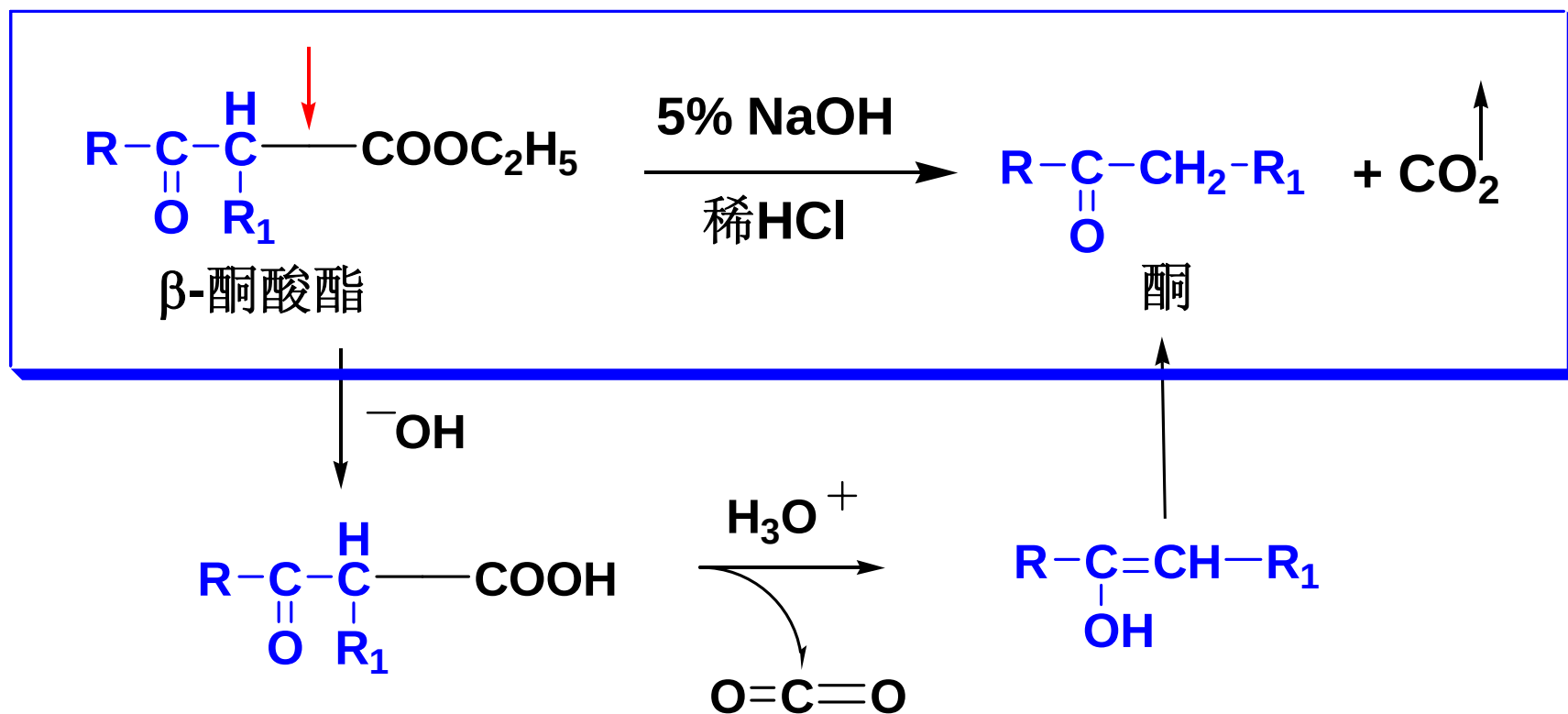
亚甲基活泼氢的性质：酸性



13.4 β -酮酸酯

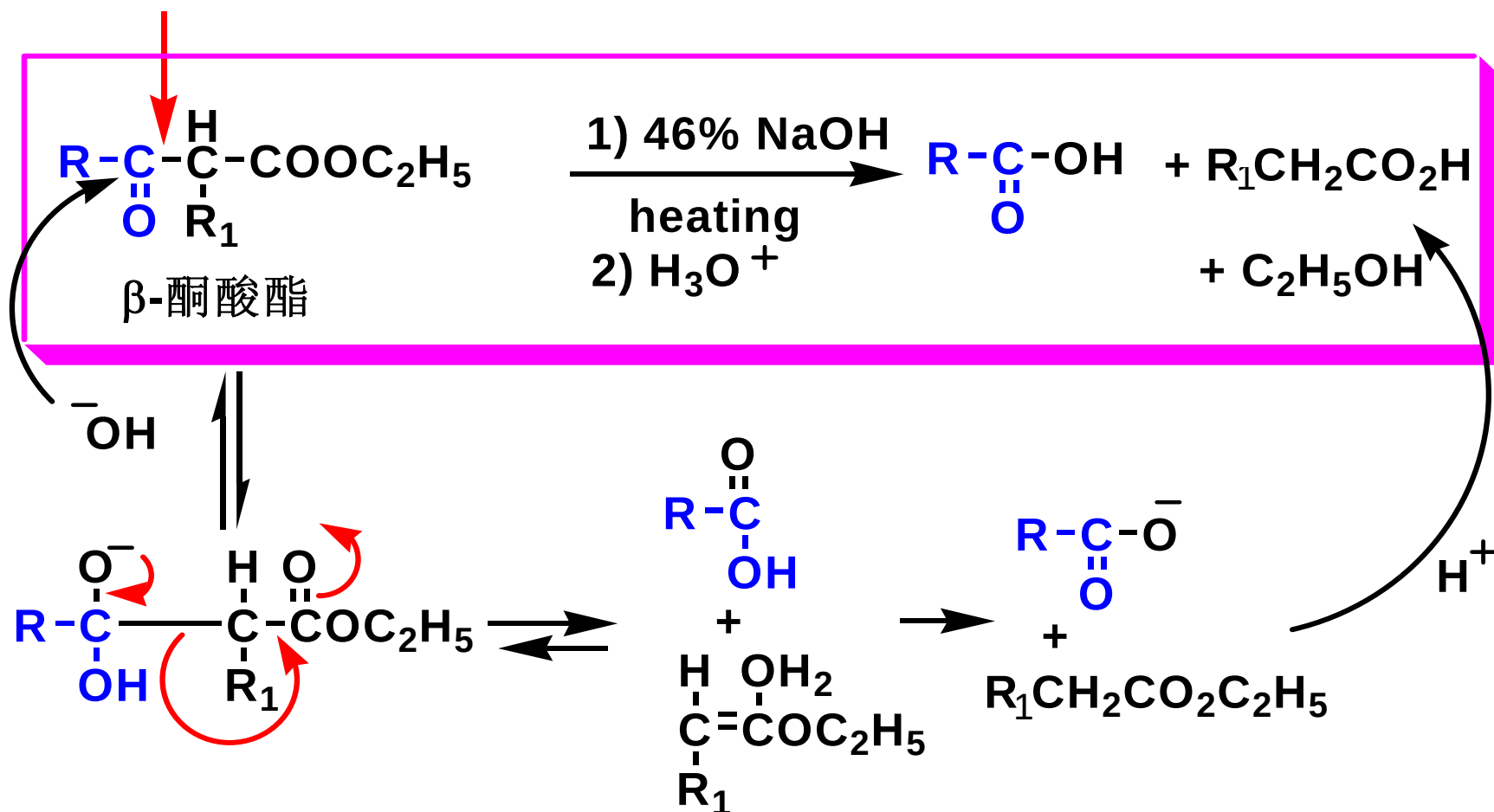
3. β -酮酸酯的水解

酮式分解:“三乙”在稀碱中水解，酸化后加热脱羧得丙酮；或者“三乙”在酸中加热脱羧得丙酮



13.4 β -酮酸酯

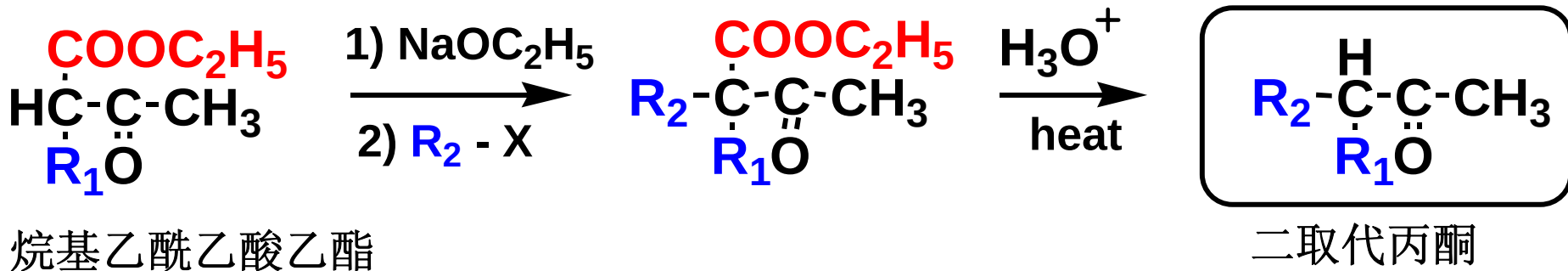
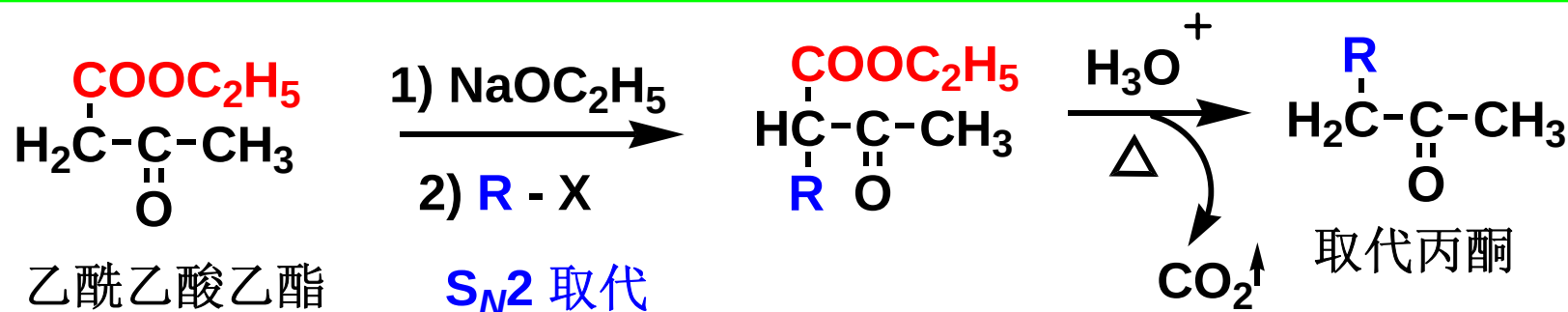
酸式分解：“三乙”在浓的强碱溶液中加热，发生酮羰基与其 α 碳键断裂，酸化后生成两个羧酸

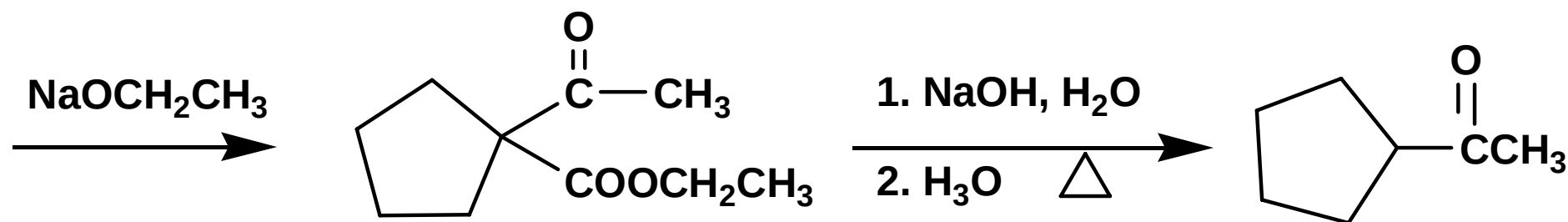
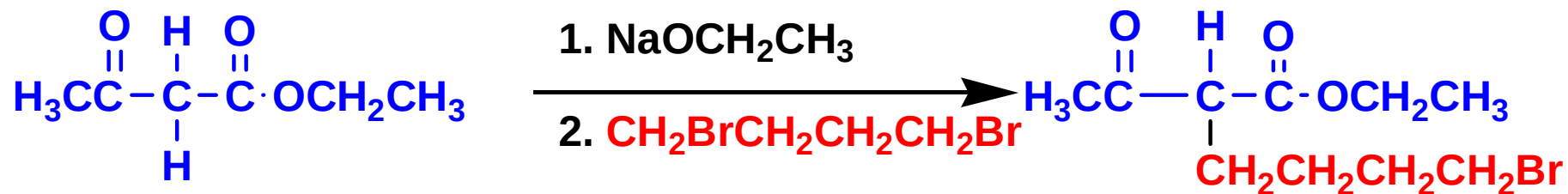


13.5 β -二羰基化合物的反应

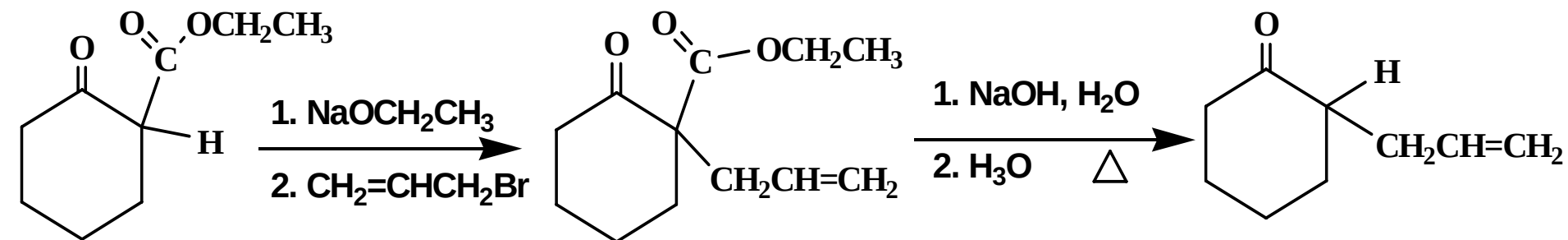
1. 乙酰乙酸乙酯在合成上的应用

合成甲基酮（掌握）

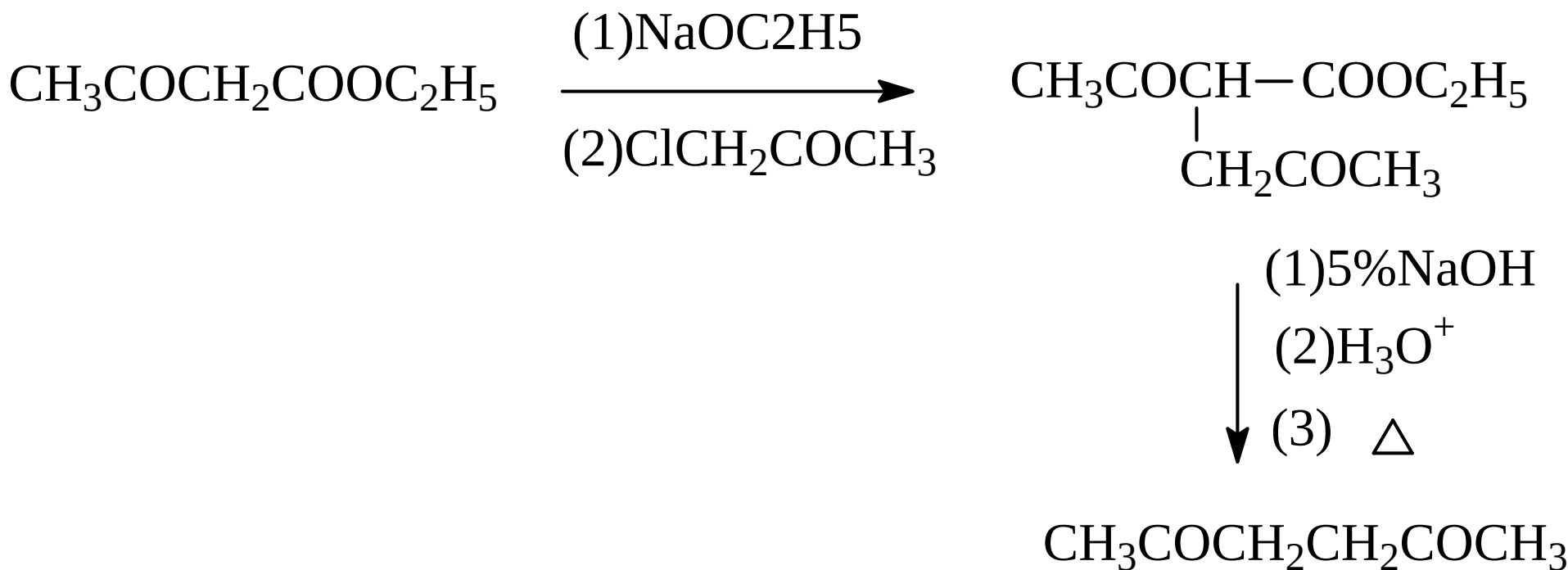




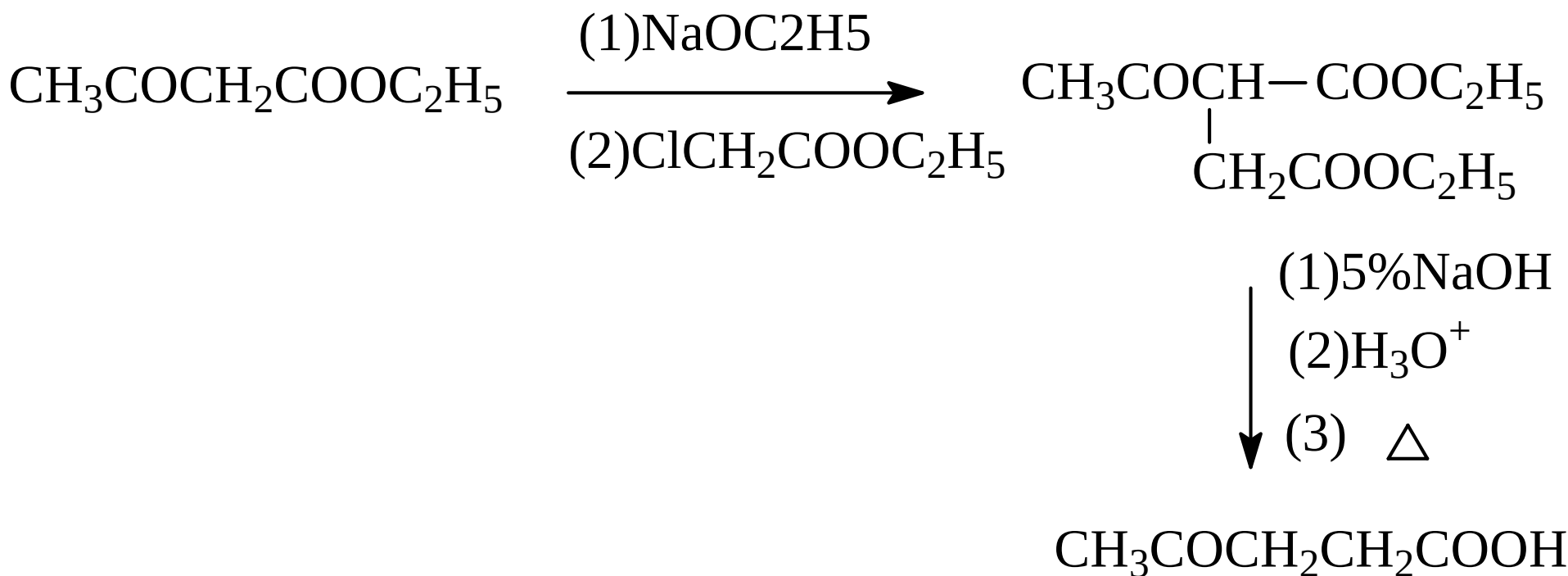
其它 β -酮酸酯也可发生类似的反应



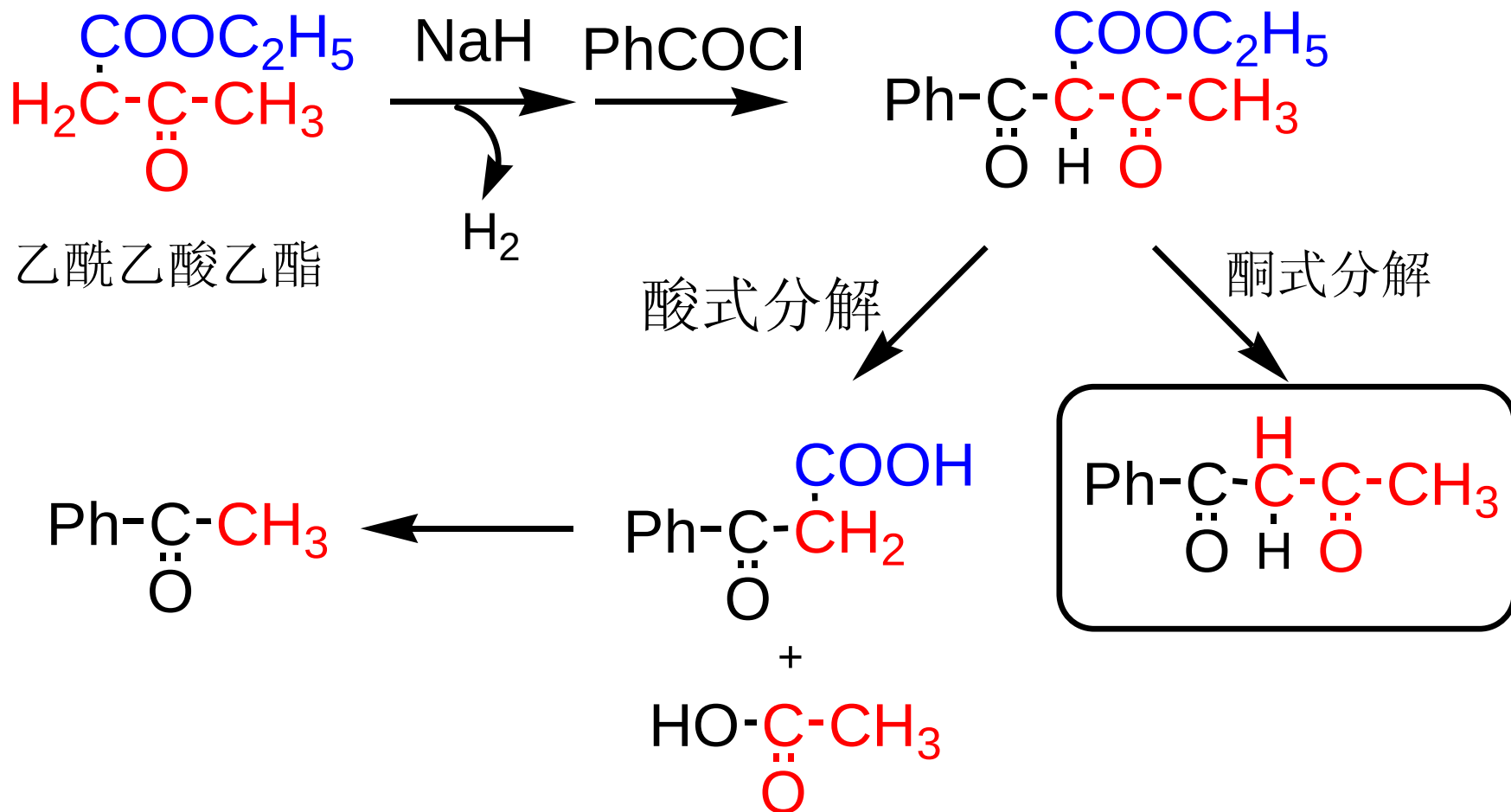
合成 γ -二酮(了解)



合成 γ -酮酸(了解)



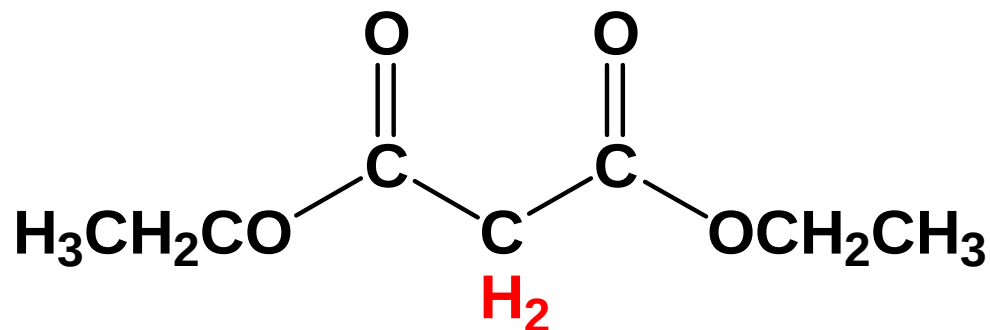
合成 β -二酮(了解)



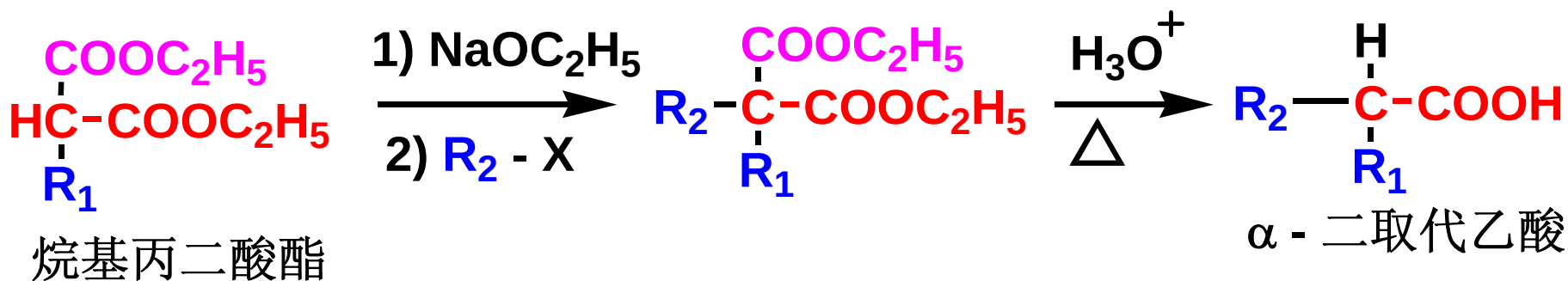
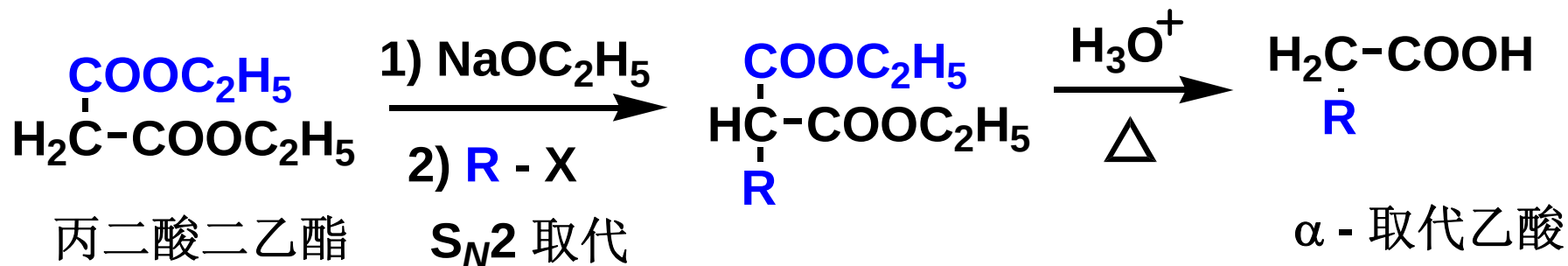
13.5 β -二羰基化合物的反应

2. 丙二酸酯在合成上的应用

丙二酸二乙酯的合成:

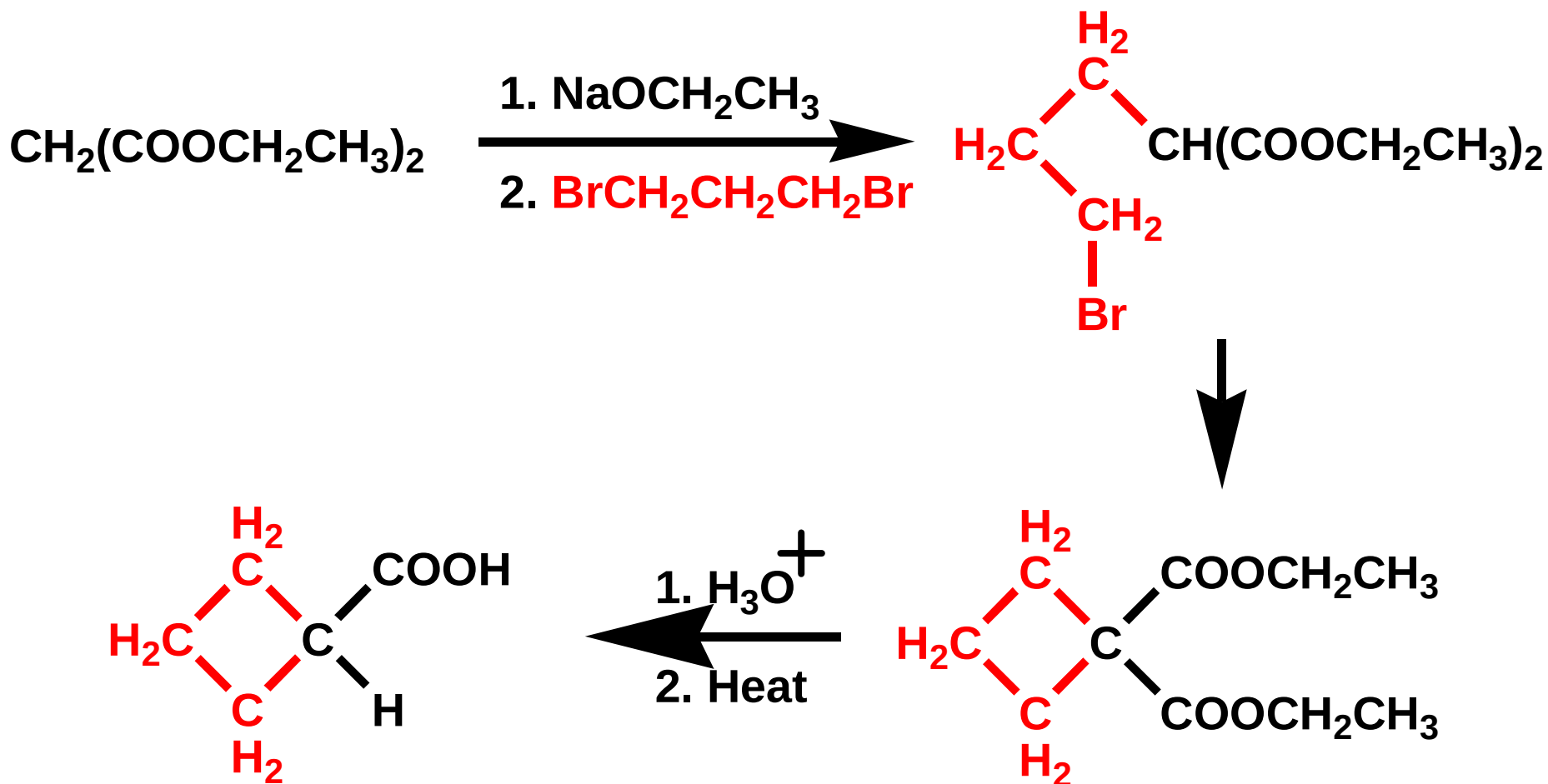


合成一元羧酸



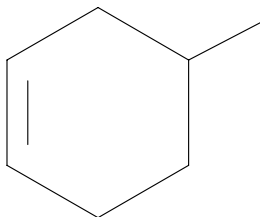
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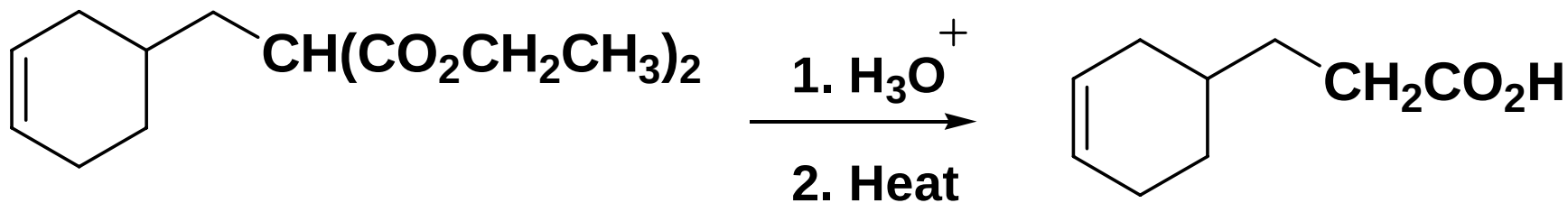
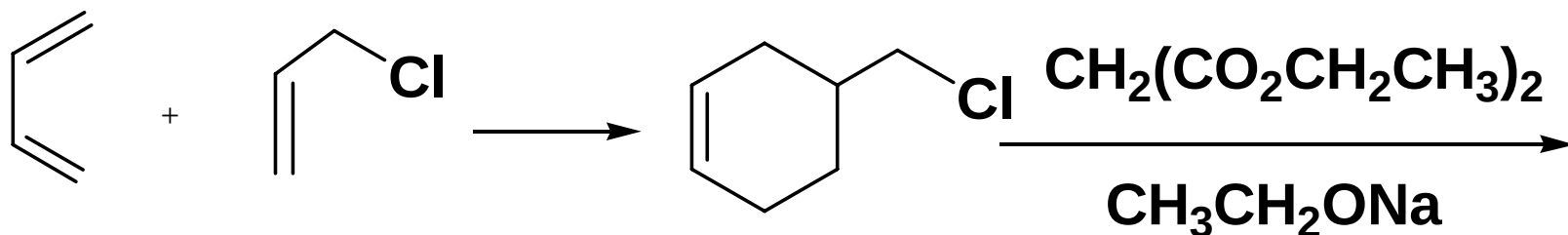
Example



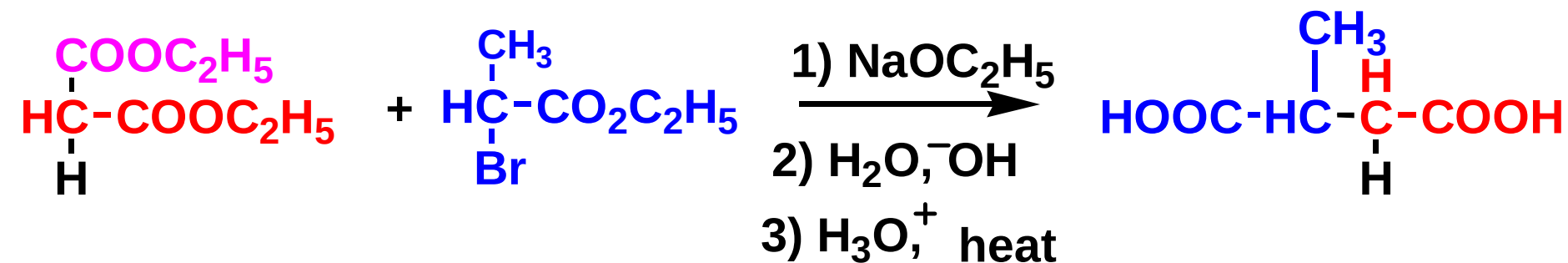
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Example

由 丙二酸酯 合成  $\text{CH}_2\text{CH}_2\text{COOH}$

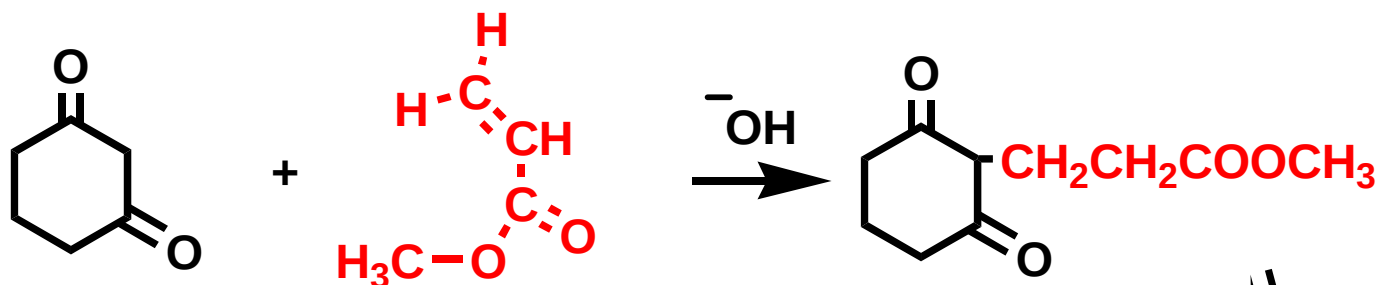


合成二元羧酸

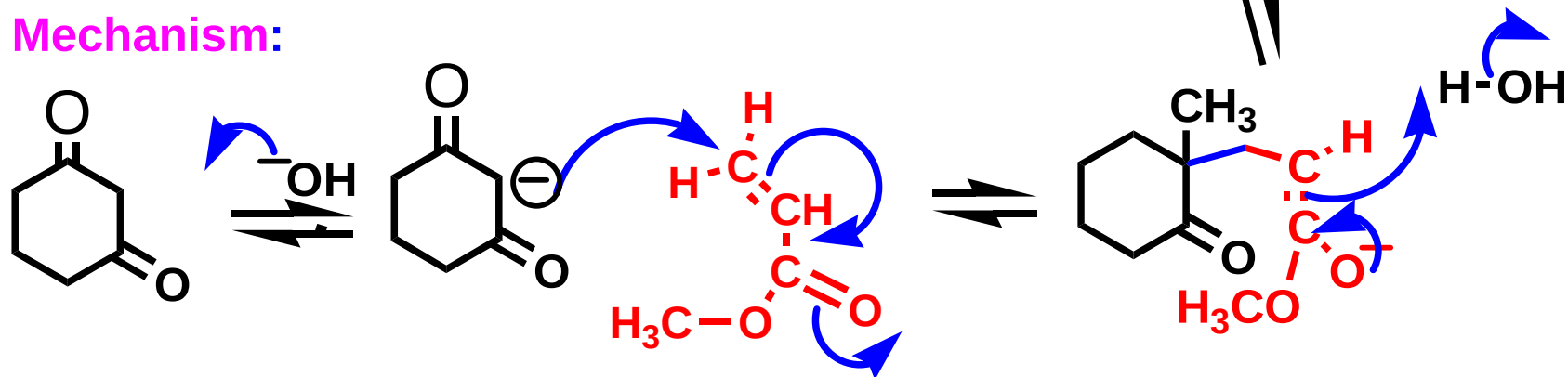


13.5 β -二羰基化合物的反应

3. Micheal 加成反应

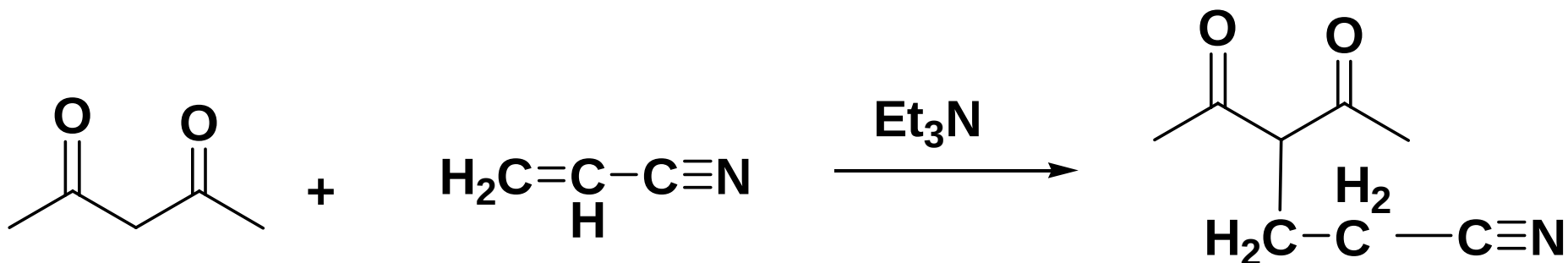
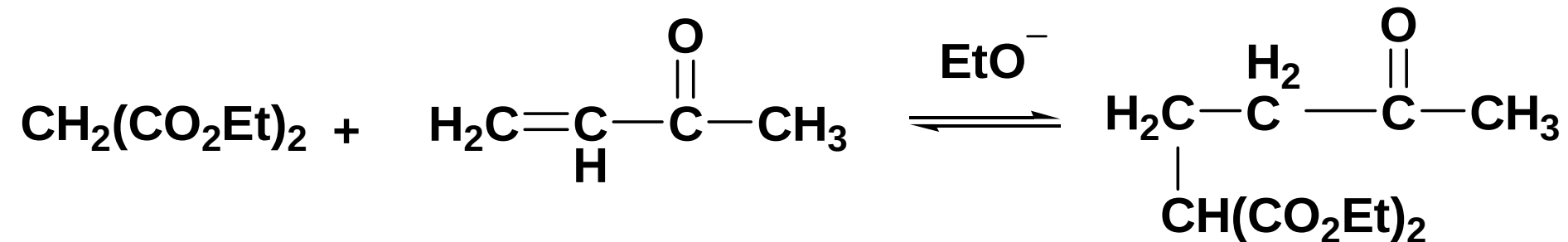


Mechanism:



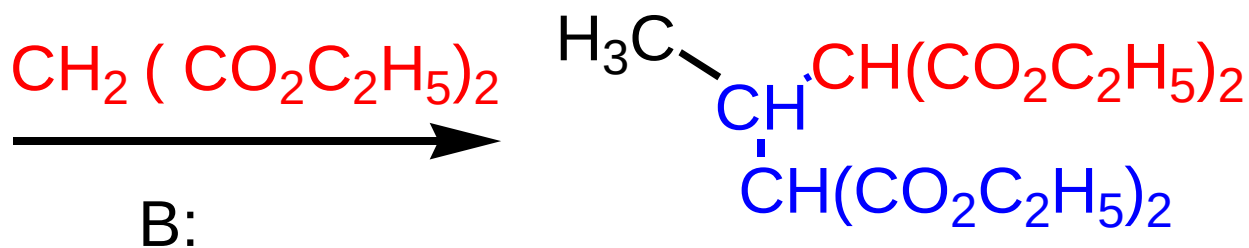
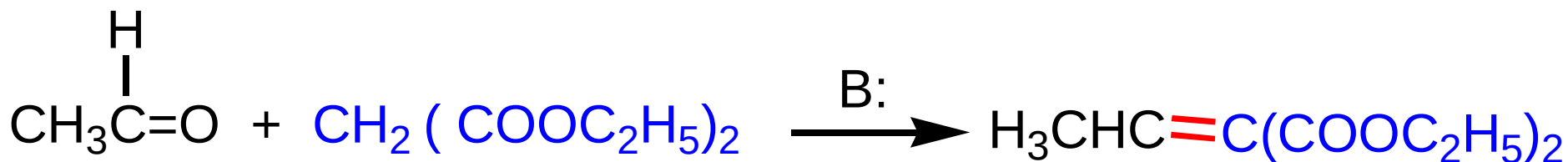
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Example



13.5 β -二羰基化合物的反应

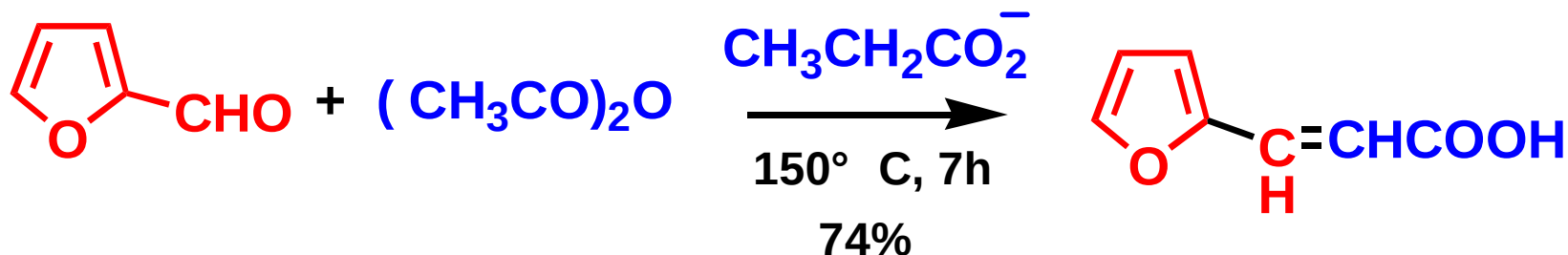
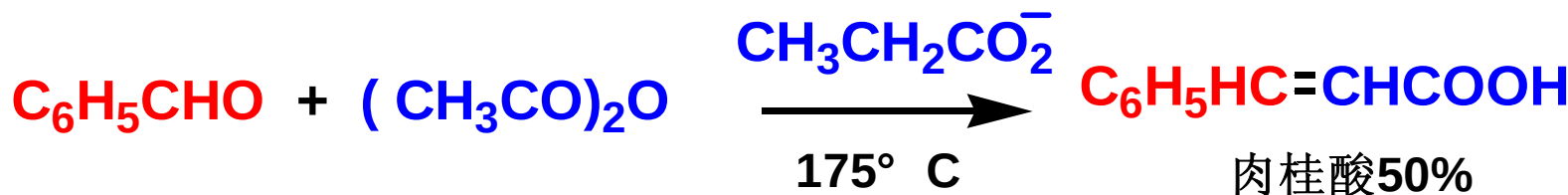
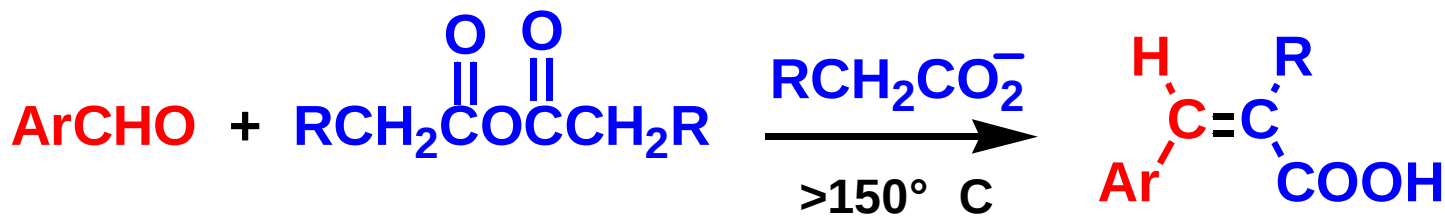
4. Knoevenagel反应



13.5 β -二羰基化合物的反应

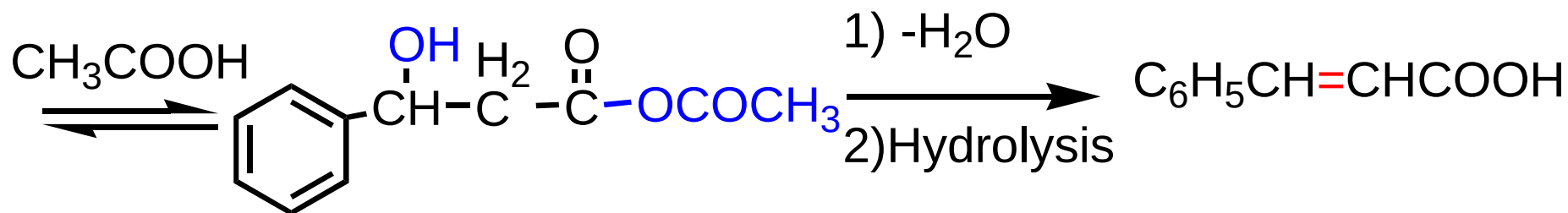
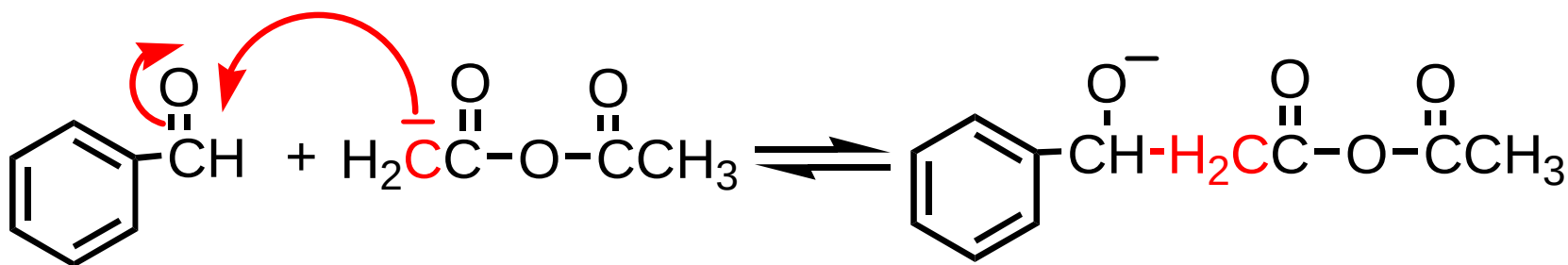
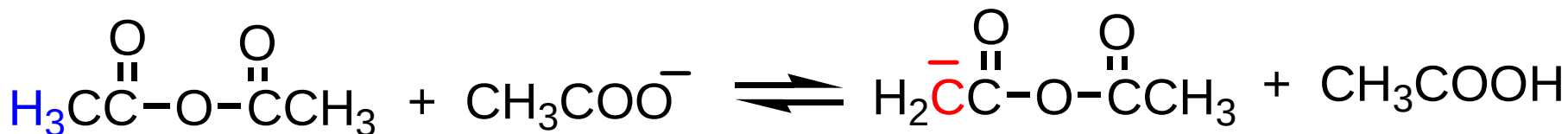
5. Perkin反应

β -芳基- α,β -不饱和酸的合成



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反应机理



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作业

- P283: 13.1 (1) ;
- P285: 13.2;
- P293: 13.5, 13.6;
- P297: 13.10, 13.11;

习题

- P306: 3, 5(1),(2), 6。